

Benefits derived by the fishermen using Potential Fishing Zone (PFZ) Advisories

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ABSTRACT

About 7 million people in India are dependent on fishing activity for their livelihood. A reliable and timely short-term forecast on the fish aggregation zones helps them. Indian National Centre for Ocean Information Services (INCOIS) provides short term forecasts using Remote Sensing (RS) and Geographic Information System (GIS) Techniques. Satellite derived Chlorophyll and Sea Surface Temperature (SST) information are the basic inputs for generating this information. Features such as oceanic fronts, meandering patterns, eddies, rings and up-welling areas are identified from the satellite images, transferred to navigational charts and provided as PFZ advisories. This paper describes the advantages resulted due to the usage of PFZ advisories that are disseminated to the fishing community of India. A quantitative analysis of the benefits in terms of net profit fetched due to the reduction in searching time and higher catch and percentage of success in the fishing operations carried out within PFZ and outside PFZ areas is presented.

Key Words: Fish, Forecast, Fishing Zones, Remote Sensing, GIS, Validation, Advisories, Chlorophyll, SST, PFZ

1. BACKGROUND

India has a coastline of 8129 km and an Exclusive Economic Zone (EEZ) of 2.02 million sq km. About 7.5 Lakhs fisher-folk families are dependent on the fishing activities. There are 58,911 Mechanized crafts, 75,591 Motorized crafts and 1,04,270 Non-motorized crafts operating in the Indian region. The annual catchable potential yield in the Indian EEZ is estimated as 3.93 mt consisting of 2.02 mt demersal, 1.67 mt pelagic and 0.24 mt oceanic resources (Anon, 2000) of which the present annual average production of about 2.55 mt forms 64.8%.

The sharing of the Pelagic and Demersal fishery resources of the continental shelf waters has created, in the recent past, considerable tensions, law and order conflicts among various fishing communities in the coastal belt. Information insufficiency on the occurrence / abundance of resources and lack of proper resource management advice are sighted as drawbacks. There is need to find a solution for the sustainable long-term economic utilisation of these resources by maintaining the exploited fish stocks through proper regulatory measures. Possibilities of bringing down the searching time for pelagic shoaling fishes, which constitute more than 55% of the total fish landings, assume great significance since this factor can bring down the cost of fishing operations by way of savings on valuable fuel oil and human effort. To achieve this, India has started a programme for issue of PFZ Advisories. INCOIS issues these PFZ Forecasts thrice a week, except during fishing ban periods and cloudy days.

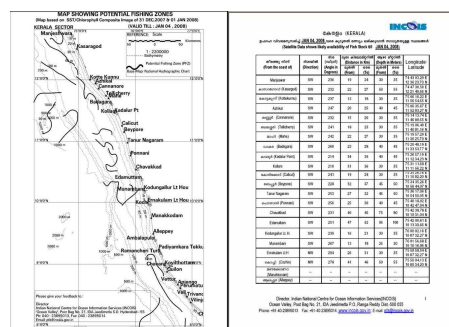
2. INTRODUCTION

It is well known that the adaptation of fish to the surrounding marine environment is controlled by various physico-chemical and biological factors. Fishes are known to react to changes in the surrounding environmental conditions and migrate to areas where favorable environmental conditions in terms of seawater temperature, salinity, dissolved oxygen levels etc., exist. Availability of food is another important factor which controls their occurrence, abundance and migrations in the sea. Sea Surface Temperature (SST) is the most easily observed environmental parameter and is quite often correlated with the availability of fish, especially pelagic fish. Changes in SST can result out of changes occurring in the direction / velocity of both horizontal and vertical circulation processes in the sea. Many pelagic species are known to concentrate at current boundaries especially in areas with sharp horizontal temperature gradients. Monitoring the above mentioned parameters in space and time is time-consuming and prohibitively expensive and a real time picture of any one of these parameters or a combination of the above becomes almost impossible. Indirect methods of

monitoring selected parameters such as SST and phytoplankton pigments (Chlorophyll-a) at sea surface from satellites is found very ideal as it provides high repetitive and synoptic coverage.

With the advent of the ocean remote sensing in eighties, fishery forecast research got much needed thrust. The linkages between satellite-derived sea surface temperature (SST) and chlorophyll with fish aggregation were established^{1, 2, 3, 4, 5}. SST and Chlorophyll over the Indian EEZ are retrieved from near-synchronous thermal infrared channels of NOAA-AVHRR (10.3 – 11.3 μm and 11.5 – 12.5 μm) and optical bands in IRS-P4 OCM data, respectively to identify PFZ along the Indian coastline and island regimes. The MCSST approach suggested by McClain⁶ is used to compute SST. Chlorophyll retrieval involves two steps, viz. atmospheric correction of visible channels of Ocean Colour Monitor (OCM) to obtain normalized water leaving radiance and application of bio-optical algorithm for retrieval of chlorophyll. The atmospheric correction of OCEANSAT-I OCM imagery is carried out using approach suggested by Mohan et al⁷. An empirical algorithm (OC2) is used to retrieve chlorophyll^{8, 9}. Both SST and chlorophyll images are corrected geometrically using a set of ground control points (GCPs) located both on the satellite image and Naval Hydrographic Office (NHO) bathymetric map. A composite image showing SST-contours on colour-coded chlorophyll image is generated, from which features such as upwelling, eddies, fronts, meanders, etc. that are important indicators for fish aggregation are identified^{10, 11, 12, 13, 14}.

These features are delineated as PFZ lines and overlaid on the base maps. These PFZ maps contain information about the major landing centres, light houses, bathymetry and latitude-longitude grids in addition to the identified fishing zones (Figure 1). For easy understanding of fishermen PFZ information is provided in both text and map format with detailed information on latitude, longitude and depth of the shelf at such locations as well as angle, direction and distance from the landing centres/light houses (Figure 1). These advisories are disseminated using phone, fax, e-mail, SMS, print media, electronic media and specially designed electronic display systems. The total number of users is estimated to be about 27,000, i. e. 27 per cent of boat owners¹⁵.



Fishing Harbours and 14 major fish landing centres from which 29,177 fishing crafts (mechanized: 5,504; motorized: 14,151 and non-motorised: 9,522) are operating for production of 5.59 Lakhs tones (2005-06) of fish landings.

4. DATA SETS AND METHODOLOGY

Potential fishing zone advisories released by INCOIS thrice a week between November and May were disseminated to active fishermen groups along the Kerala coast between Vizhinjam in the south and Cannore in the north by fax/telephonic messages and digital display boards installed at five major fish landing centres. The feedback data is gathered in the prescribed format of INCOIS immediately on arrival of the vessel at the landing centre on completion of fishing activity. The data gathered using the feedback form such as (i) location of fishing, (ii) type of craft, (iii) type of gear, (iv) fishing operation details, (v) catch details and (vi) expenditure details are used to estimate the usefulness of PFZ advisories.

4.1. Calculation of percentage reduction in searching time and resulting savings on fuel consumption for individual fishing operations.

A total of 87 PFZ advisories generated by INCOIS between October, 2003 and May, 2005 were validated by collecting 1500 feedback datasets from 295 gill netters and 400 ring seiners within and outside the notified PFZ locations. The time taken by the vessel to reach the shoal with the help of PFZ advisory and in the absence of an advisory (for the same area/season/targeted species/type of boat/size of boat/engine horse power/human effort) is compared before arriving at the actual advantage derived. The changes in advantage also varies with the type/size/engine horsepower of the craft, size of gear, total human effort employed, area/season of fishing activity and above all the species targeted. The advantage derived is also dependent on the cost of fuel oil (HSD/kerosene) at the point of time of fishing activity. Details of the advisories used for the study are given in Table 1.

4.2. Calculation of percentage improvement in fish catch and resulting increase in net income for individual fishing operations.

A total of 18 PFZ advisories generated by INCOIS between April 2006 and February 2008 were validated by conducting controlled experiments utilizing more or less identical fishing vessels, fishing gear and fishing duration in which simultaneous fishing operations were conducted within and outside PFZ area.

4.3. Calculation of percentage success of operations for different types of vessels within and outside PFZ areas:

A total of 18 PFZ advisories generated by INCOIS between April 2006 and February 2008 were validated by using collateral data collected from 236 Ring Seiners, 125 Gill Netters and 65 Bottom Trawlers, operating both within and outside PFZ locations. While calculating the percentage of success, a fishing operation has been considered successful if a net profit in excess of 10 % of total expenditure is recorded. Details of the advisories used for the study are given in Table 2.

Table 1: Number of PFZ Advisories validated for the present study

Mon/Year	(1)	(2)	(3)	Mon/Year	(1)	(2)	(3)
Oct 2003	1	11	7	Nov 2004	2	21	5
Nov 2003	9	115	16	Dec 2004	9	150	23
Dec 2003	9	103	13	Jan 2005	4	154	44
Jan 2004	9	69	13	Feb 2005	7	206	62
Feb 2004	7	77	18	Mar 2005	10	200	40
Mar 2004	6	97	18	Apr 2005	5	90	26
Apr 2004	4	98	17	May 2005	4	82	30
May 2004	1	24	20	Total	87	1497	357

(1) PFZ advisories validated, (2) Total No. of fishing vessels covered and (3) Total No. of landing centers covered

Table 2: Number of PFZ Advisories validated for the present study

Expt. No.	Place and Date	Fishing technique	No of vessels (Within PFZ + Outside PFZ)
1	Vypeen; 15.04.2006	Ring Seining	16(11+5)
2	Neendakara; 11.05.2006	Ring Seining	20(10+10)
3	Vypeen; 16.12.2006	Ring Seining	18(12+6)
4	Puthiappa; 24.01.2007	Ring Seining	17(9+8)
5	Puthiappa; 24.02.2007	Ring Seining	19(9+10)
6	Chavakkad; 22.03.2007	Ring Seining Bottom Trawling	12(9+3) 8(8+0)
7	Puthiappa; 20.04.2007	Ring Seining Gillnetting	9 (5+4) 4 (4+0)
8	Kovilhottam; 05.05.2007	Ring Seining Gillnetting	7 (1+6) 8 (3+5)
9	Ramancherihure; 24.08.2007	Ring Seining Gillnetting Bottom Trawling	24 (4+20) 24 (8+16) 23(12+11)
10	Kalamukku; 24.08.2007	Ring Seining Gillnetting	11(3+8) 18(9+9)
11	Kalamukku; 31.08.2007	Ring Seining Gillnetting Bottom Trawling	16(9+7) 14(7+7) 8(4+4)
12	Kovilhottam; 01.10.2007	Ring Seining Gillnetting	14(6+8) 12(6+6)
13	Puthiappa; 29.11.2007	Ring Seining Gillnetting	10(5+5) 10(5+5)
14	Padiamkarathekku; 21.12.2007	Ring Seining Gillnetting Bottom Trawling	14(6+8) 8(4+4) 4 (4+0)
15	Padiamkarathekku; 09.01.2008	Ring Seining Gillnetting Bottom Trawling	6(2+4) 5(3+2) 5(3+2)
16	Mankkodam; 11.01.2008	Ring Seining Gillnetting Bottom Trawling	5(1+4) 4(3+1) 5(3+2)
17	Puthiappa; 02.02.2008	Ring Seining Gillnetting Bottom Trawling	12(6+6) 10(5+5) 12(6+6)
18	Padiamkarathekku; 13.02.2008	Ring Seining Gillnetting	6(3+3) 8(4+4)
Total		Ring Seining Gill Netting Bottom Trawling	236(119 + 117) 125(61+ 64) 65(40+25)

5. RESULTS AND DISCUSSION

Positive relationship between PFZ resulting out of comparatively high gradients of SST and Chlorophyll and fishable concentrations of commercially important fishes was found in respect of pelagic and column fishing activities such as purse seining, ring seining, gill netting and trolling. SST based advisories were found more advantageous for locating

mackerel, tuna, anchovies and carangids whereas the chlorophyll based advisories were found more advantageous for locating oil sardine and lesser sardine shoals which are herbivorous in nature with feeding preferences for phytoplankton. Comparison of reduction in searching time for oil sardines obtained for PFZ forecasts originating out of SST images and chlorophyll images revealed the advantage of the latter for locating oil sardine shoals. SST based images were also found useful for locating concentration of young ones of yellow fin tuna (popularly known as “Kera”).

In order to make profit, individual boat will have to reduce the cost of operation to the barest minimum by way of reducing the fuel consumption since the actual fish catch will depend on many environmental factors which influence the availability and abundance of a particular species of fish in space and time. In the case of fishing targeted at a pelagic resource which is migratory in habitat, the PFZ will definitely provide an advantage in terms of reduction in searching time, which translates as reduction in the consumption of fuel oil and human effort there by bringing down the total cost of fishing operations.

5.1. Quantification of the reduction in search time and resultant saving in diesel consumption:

Maximum advantages in terms of reduction in searching time was found in respect of ring seining and purse seining of fishes like oil sardine, anchovies, coastal tunas and young ones of oceanic tunas, carangids and seer fish. In the case of mackerel, the advantages were found more for purse-seining since the matured fish is known to undertake vertical migrations. Reduction in searching time is found to be maximum (60-70%) for oil sardine shoals in ring seining with 30-50% reduction reported for mackerel, anchovy, tuna and carangid shoals in ring seining and purse seining operations. The reduction in searching time reported for pelagic shoaling fishes is mainly dependent on three factors viz. (i) time lag between satellite data acquisition and actual fishing time, (ii) fishing method adopted viz., gill netting, purse seining, ring seining and trolling as well as time at which the fishing activity is undertaken and (iii) type of fish targeted.

From the data analysed, the quantitative advantage expressed as per cent gain varied between 30% and 70% (avg.50%). A modest computation of quantitative advantage derived through the usage of PFZ advisories along the Kerala coast for different types of vessels considering an average of 50% reduction in search time indicates the following:

- In the case of a large sized inboard motor vessel (mechanised) engaged in ring seining (with engine HP 100-106) the diesel consumption would be about 18 l/hour and works out to INR.6 lakhs (one lakh = 100,000 ; INR one lakh = eq. USD 2200) for a 6 month period (when PFZ advisories are utilised by the vessel). The basis for this calculation is given in Table 3

Table 3: Details of the quantitative advantage derived by fishing vessels using PFZ Advisories

1.	Diesel Consumption /hr (average)	18 litres
2.	Total requirement of HSD for reaching a fish shoal located 30 km away from landing centre (to and fro) + for encircling calculated at a modest speed of 10km/hr for 7 hours (6 hrs for steaming and 1 hour for encircling the shoal)	18 x 7 = 126 litres (say 130 litres)
3.	Calculating at the rate of INR.30/litre of HSD, the cost of fuel oil works out to	130 x 30 = INR.3900
4.	+ Other lubricants	INR.100
5.	Total	INR.4000/day of operation
6.	Cost of HSD for a week (considering 6 days operation)	INR.24,000/-
7.	Cost of HSD for 25 weeks operation between November and April in the case of vessels utilising PFZ advisories – considering an average of 50% reduction in searching time and proportionate reduction in the consumption of fuel oil (as reported by fishermen/fishing vessel operators)	INR.6 lakhs
8.	Cost of HSD for 25 weeks operation between November and April in the case of vessels not utilising PFZ advisories (as reported by fishermen/fishing vessel operators from their past experience)	INR.12 lakhs
9.	Quantitative advantage derived by individual boat for 6 month period (in terms of 50% reduction in the consumption of fuel oil)	INR.6 lakhs

- In the case of ring seiners fitted with outboard motors of 40 HP, the consumption will work out to 10 l/hr., and the advantage would be around INR.1.95 lakhs for a 6-month period, considering the limitations in respect of area of operation and size of fishing gear.
- In the case of gill netters with outboard motors 9.9 HP / 15 HP / 25 HP the average consumption of kerosene will work out to 8 litres of kerosene per hour and calculating at a unit selling price of kerosene at INR.30/lit, the advantage derived would be around 1.56 lakhs per boat for a six month period, considering the limitations in respect of area of operation and the passive nature of gill netting activity.

As per INCOIS estimates, 27 % of the boat owners are using PFZ advisories. As per estimates of CMFRI, Kerala has 5,504 mechanised vessels, 14,151 motorised vessels and 9,522 non-motorised vessels. Considering that 27 % of the boat owners are using PFZ advisories that amounts to about 1,486 mechanised boats, 3,820 motorised boats and 2,571 non-motorised boats. Based on conservative estimates from the current study, this amounts to an annual savings to the tune of Rs.89 Crores on account of diesel savings for all the mechanised boats that are using PFZ advisories and Rs.74 Crores for motorised boats. This totals to an annual savings of Rs.163 crores for Kerala coast. If 100 % of the mechanised and motorised boats operating in Kerala use PFZ advisories, this will account for an annual savings of about Rs.600 Crores just on account of diesel savings in addition to the valuable human effort.

An attempt was made to extrapolate the above results to derive a national perspective of the benefits derived due to usage of PFZ advisories. These calculations take into account the total number of mechanised and motorised boats in the country. As per estimates of CMFRI, there are 58,911 Mechanised crafts and 75,591 Motorised crafts in the Country. Considering that 27 % of the boat owners are using PFZ advisories this amounts to about 15,900 mechanised boats and 20,400 motorised boats. Based on conservative estimates from the current study, this amounts to an annual savings to the tune of Rs.954 Crores on account of diesel savings for all the mechanised boats that are using PFZ advisories and Rs.397 crores for motorised boats. This totals to an annual savings of Rs.1351 crores for Indian coast. This estimate closely matches with that estimated by the Madras School of Economics. If 100 % of the mechanised and motorised boats operating in India use PFZ advisories, this will account for annual savings of about Rs.5000 Crores just on account of diesel savings in addition to the valuable human effort. Putting it in a different perspective, the national consumption of diesel on this count would reduce by 166 Cr litres per annum. This will go a long way in even reducing environmental pollution.

5.2. Quantification of increase in fish catch in PFZ areas and resultant increase in income

To arrive at one-on-one comparison between fish catch in the notified and non-notified location, sufficient care was taken to ensure that same type of fishing vessel and fishing gear was used for conducting simultaneous fishing in both the locations. All the 18 experiments were pelagic fishing activities and were conducted using ring seiners. A sample map of one of the experiments showing the location of actual fishing overlaid on PFZ Advisory and corresponding economics of fishing operations are given in Figure 2.

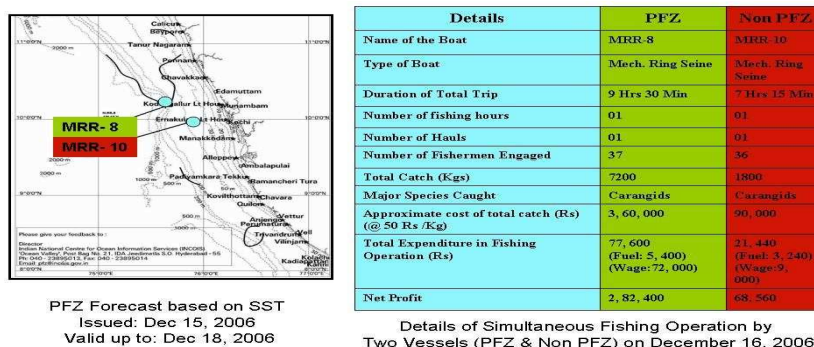


Fig. 2: Location Map and corresponding economics of fishing operations for controlled experiment 3 conducted on Dec 16, 2006

The actual fishing effort was limited to one hour in each case and hence CPUE as indicated in these experiments indicates the actual quantum of catch from each fishing operation. The economics of fishing operation were estimated for each fishing operation taking into account, the income earned by sale of the catch and the expenses incurred on account of fuel and wages. The results of simultaneous fishing operations in notified PFZ and non-notified areas are given in the Table 4.

Table 4: Results of simultaneous fishing operations in PFZ and Non-PFZ areas

Exp t. No.	Location	Major species reported in the Catch	CPUE (Kg)	Revenue earned from the fish catch (Rs.)	Total Expenditure (Rs.)	Net Profit (Rs.)
1	Vypeen 15.04.2006 Ring Seining					
	Within PFZ	Oil Sardine	6,500	10,4000	46,200	57,800
	Outside PFZ	Oil Sardine	1,600	2,5600	13,640	11,960
2	Neendakara 11.05.2006 Ring Seining					
	Within PFZ	Coastal Tuna	5,800	1,74,000	9,340	82,660
	Outside PFZ	Coastal Tuna	1,100	33,000	20,320	12,680
3	Vypeen 16.12.2006 Ring Seining					
	Within PFZ	Carangids	7,200	3,60,000	77,600	2,82,400
	Outside PFZ	Carangids	1,800	90,000	12,400	77,560
4	Puthappa 24.01.2007 Ring Seining					
	Within PFZ	Coastal Tuna	4,100	2,46,000	1,28,960	1,17,040
	Outside PFZ	Coastal Tuna	850	51,000	30,740	20,260
5	Puthappa 24.02.2007 Ring Seining					
	Within PFZ	Coastal Tuna	3,800	1,90,000	99,820	90,000
	Outside PFZ	Coastal Tuna	700	35,000	23,040	12,000
6	Chavakkad 22.03.2007 Ring Seining					
	Within PFZ	Mackerel	1,800	81,000	46,100	40,500
	Outside PFZ	Mackerel	700	31,500	21,700	15,750
7	Puthappa 20.04.2007 Ring Seining					
	Within PFZ	Mackerel	1,300	52,000	5,481	46,519
	Outside PFZ	Mackerel	275	11,000	3,980	7,020
8	Kovilthottam 05.05.2007 Ring Seining					
	Within PFZ	Horse Mackerel	4,100	1,23,000	67,380	55,620
	Outside PFZ	Horse Mackerel	900	29,000	17,600	9,400
9	Ramancherihure 4.08.2007 Ring Seining					
	Within PFZ	Mackerel	4,700	1,88,000	96,990	91,910
	Outside PFZ	Mackerel	820	32,800	18,234	14,566
10	Kalamukku 24.08.2007 Ring Seining					
	Within PFZ	Oil Sardine	3,000	75,000	40,490	34,510
	Outside PFZ	Oil Sardine	1,000	25,000	14,912	10,088
11	Kalamukku 31.08.2007 Ring Seining					
	Within PFZ	Mackerel	2,500	1,00,000	54,146	45,854
	Outside PFZ	Mackerel	625	25,000	17,224	7,776
12	Kovilthottam 29.09.2007 Ring Seining					
	Within PFZ	Coastal Tuna	3,800	1,14,000	59,412	54,588
	Outside PFZ	Coastal Tuna	680	20,400	13,190	7,210
13	Puthiappa 30.11.2007 Ring Seining					
	Within PFZ	Oil Sardine	3,300	82,500	45,400	37,100
	Outside PFZ	Oil Sardine	600	15,000	10,500	4,500

14	Padiamkarathekku 21.12.2007 Ring Seining					
	Within PFZ	Mackerel	1,800	90,000	47,700	42,300
	Outside PFZ	Mackerel	600	30,000	18,840	11,160
15	Padiamkarathekku 09.01.2008 Ring Seining					
	Within PFZ	Horse Mackerel	2,900	87,000	46,500	40,500
	Outside PFZ	Horse Mackerel	600	29,400	17,700	11,700
16	Mankkodam 10.01.2008 Ring Seining					
	Within PFZ	Mackerel	1,950	97,500	52,318	45,187
	Outside PFZ	Mackerel	550	24,750	15,375	9,375
17	Puthiappa 02.02.2008 Ring Seining					
	Within PFZ	Horse Mackerel	3,200	96,000	51,600	44,400
	Outside PFZ	Horse Mackerel	725	21,750	14,475	7,275
18	Padiamkarathekku 12.02.2008 Ring Seining					
	Within PFZ	Oceanic Tuna	600	60,000	2,600	27,400
	Outside PFZ	Oceanic Tuna	150	15,000	1,100	6,400

Increase in CPUE: Data collected from 18 controlled experiments indicated that the Catch Per Unit Effort (CPUE) for the vessels that conducted fishing operations in PFZ locations were consistently higher than those compared to the vessels fishing in the non notified locations. The average CPUE for vessels fishing within the notified PFZ locations was about 3464 Kgs as against an average of 793 Kgs for vessels fishing outside the PFZ locations. This means that the average catch in notified locations is about 4.3 times higher than that obtained from the non notified locations. Experiment wise CPUE is shown in Figure 3 (a).

Increase in net Profit: The economics of fishing operation were estimated for each fishing operation taking into account, the income earned by sale of the catch and the expenses incurred on account of fuel and wages. In case of all the 18 experiments, the income from the vessels fishing in notified PFZ locations was consistently higher than those compared to the vessels fishing in the non-notified locations. The average net profit of vessels fishing within the notified PFZ locations was Rs.68,683 as against an average of Rs.14,260 for vessels fishing outside the PFZ locations. This means that the average net profit in notified locations is about 4.8 times higher than that obtained from the non-notified locations. The increase in net profit is a direct result of more fish catch reported in the notified PFZ locations. Experiment wise Net Profit of fishing operations were shown in Figure 3 (b).

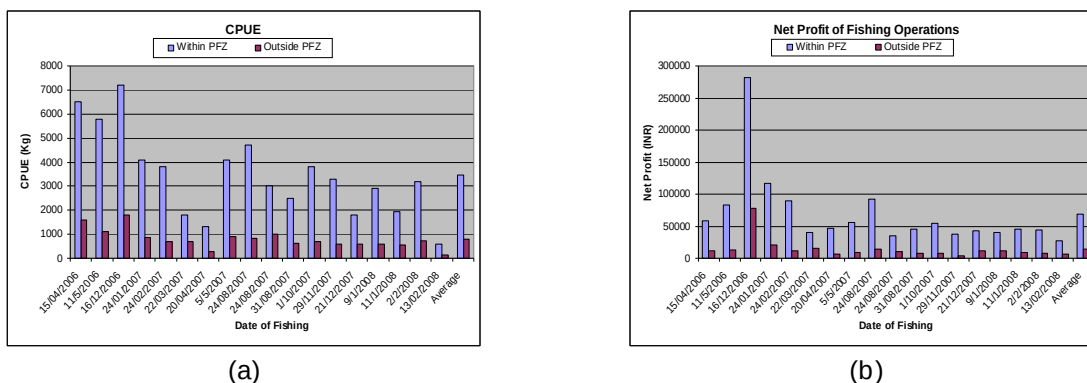


Fig. 3: (a) Catch Per Unit Effort and (b) Net profit earned by the fishing operations done within PFZ and Outside PFZ Locations

5.3. Quantification of Percentage Success within PFZ and outside PFZ areas

To estimate the Percentage success of PFZ advisories, feedback data was collected from vessels operating in the notified PFZ and non-notified areas and validated against 18 PFZ advisories issued by INCOIS from April 2006 to February

2008. During this period, data was collected from 236 ring seiners, 125 gill netters and 65 bottom trawlers fishing in both notified PFZ and non-notified areas. While calculating the percentage of success, a fishing operation has been considered successful if a net profit in excess of 10 % of total expenditure is recorded. The results of simultaneous fishing operations in notified PFZ locations and non-notified locations are given in the Table 5.

Table 5: Percentage of Success within notified PFZ and non-notified locations

Expt. No.	Place and Date of conducting the experiment	Fishing technique employed	Major species caught	No of vessels (Within PFZ + Outside PFZ)	Percentage of Success	
					With in PFZ	Out side PFZ
1	Vypeen, 15.04.2006	Ring Seining	Oil sardine	16(11+5)	82	20
2	Neendakara, 11.05.2006	Ring Seining	Coastal Tuna	20(10+10)	60	20
3	Vypeen, 16.12.2006	Ring Seining	Carangids	18(12+6)	83	30
4	Puthappa, 24.01.2007	Ring Seining	Young Oceanic Tuna	17(9+8)	100	37
5	Puthiappa, 24.02.2007	Ring Seining	Young Oceanic Tuna	19(9+10)	89	40
6	Chavakkad, 22.03.2007	Ring Seining	Mackerel	12(9+3)	77	33
		Bottom Trawling	Mackerel	8(8+0)	50	--
7	Puthappa, 20.04.2007	Ring Seining	Mackerel	9 (5+4)	100	25
		Gillnetting	Mackerel	4 (4+0)	100	--
8	Kovilhottam, 05.05.2007	Ring Seining	Horse Mackerel	7 (1+6)	100	0
		Gillnetting	Horse Mackerel	8 (3+5)	100	0
9	Ramancherihure, 24.08.2007	Ring Seining	Mackerel	24 (4+20)	100	5
		Gillnetting	Mackerel	24 (8+16)	100	0
		Bottom Trawling	Mackerel	23(12+11)	100	0
10	Kalamukku, 24.08.2007	Ring Seining	Oil sardine	11(3+8)	100	75
		Gillnetting	Oil sardine	18(9+9)	100	22
11	Kalamukku, 31.08.2007	Ring Seining	Mackerel	16(9+7)	100	12
		Gillnetting	Mackerel	14(7+7)	100	14
		Bottom Trawling	Mackerel	8(4+4)	75	0
12	Kovilhottam, 01.10.2007	Ring Seining	Coastal Tuna	14(6+8)	100	12
		Gillnetting	Coastal Tuna	12(6+6)	83	16
13	Puthiappa, 29.11.2007	Ring Seining	Oil sardine	10(5+5)	60	0
		Gillnetting	Oil sardine	10(5+5)	100	0
14	Padiamkarathekku, 21.12.2007	Ring Seining	Mackerel	14(6+8)	100	12
		Gillnetting	Mackerel	8(4+4)	100	75
		Bottom Trawling	Mackerel	4 (4+0)	75	--
15	Padiamkarathekku, 09.01.2008	Ring Seining	Horse Mackerel	6(2+4)	100	75
		Gillnetting	Horse Mackerel	5(3+2)	75	0
		Bottom Trawling	Horse Mackerel	5(3+2)	75	0
16	Mankkodam, 11.01.2008	Ring Seining	Mackerel	5(1+4)	100	0
		Gillnetting	Mackerel	4(3+1)	100	0
		Bottom Trawling	Mackerel	5(3+2)	100	0
17	Puthiappa, 02.02.2008	Ring Seining	Horse Mackerel	12(6+6)	100	66
		Gillnetting	Horse Mackerel	10(5+5)	80	0
		Bottom Trawling	Horse Mackerel	12(6+6)	83	0
18	Padiamkarathekku, 13.02.2008	Ring Seining	Coastal Tuna	6(3+3)	100	60
		Gillnetting	Oil sardine, Horse Mackerel	8(4+4)	100	0
Total		Ring Seining Gill Netting Bottom Trawling		236(119 + 117) 125(61+ 64) 65(40+25)		

Data collected from 236 ring seiners indicated that the average success rate of fishing operations within the PFZ locations is about 92% as compared to the success rate of 29% outside the PFZ Locations. On 12 out of the 18 occasions of ring seining, 100% of the vessels that operated with in the PFZ locations reported successful operations. Details are given in Figure 4 (a). Whereas the data collected from 125 gill netters indicated that the average success rate of fishing operations within the PFZ locations is about 95% when compared to the success rate of 12% outside the PFZ Locations. On 09 out of the 12 occasions of gill netting, 100% of the vessels that operated with in the PFZ locations reported successful operations. Details are given in Figure 4 (b). In the case of Bottom Trawling Operations, the data collected from 65 bottom trawlers indicated that the average success rate of fishing operations within the PFZ locations is about 85% when compared to the success rate of 0% outside the PFZ Locations. Only on 02 out of the 07 occasions of bottom trawling, 100% of the vessels that operated with in the PFZ locations reported successful operations. Details are given in Figure 4 (c).

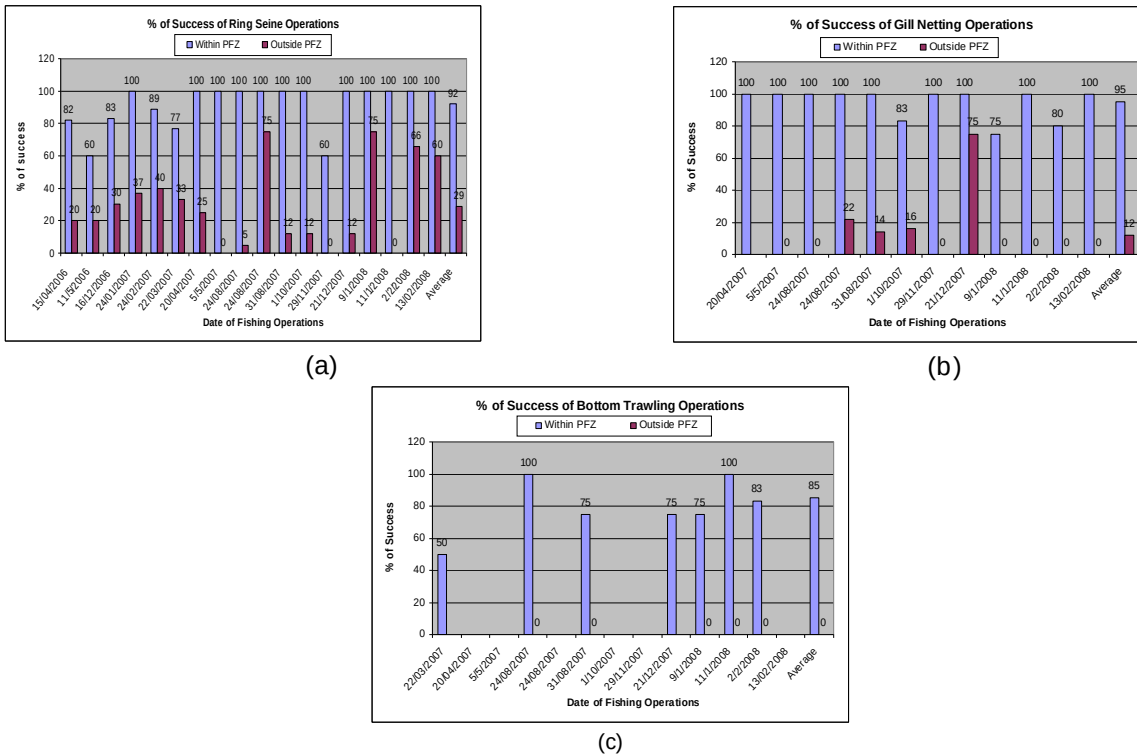


Fig. 4: Percentage of Success of fishing operations within PFZ and Outside PFZ Locations for (a) Ring Seining (b) Gill Netting and (c) Bottom Trawling

To sum up, percentage success of fishing operations in the notified PFZ locations was 92 % for ring seining, 95 % for gill netting and 85 % for bottom trawling operations as compared to a meager 29 %, 12 % and 0 % respectively in the non-notified locations. These experiments conclusively prove the effectiveness of using Potential Fishing Zone advisories.

6. SUMMARY AND CONCLUSIONS

PFZ advisories generated from satellite retrieved SST and chlorophyll were found to be good indicators of the availability/abundance of pelagic fishes such as sardines, mackerel, anchovies, tunas, carangids etc. Hence these advisories are found beneficial to artisanal, motorised and small mechanised sector fishermen engaged in pelagic fishing

activities such as ring seining, purse seining, gill netting etc., for easily locating concentrations of these highly migratory pelagic species, thereby reducing the searching time and in turn saving valuable fuel oil which is becoming dearer day by day. A summary of the results is given below:

Reduction in Search time and consequent savings on diesel consumption for different types of vessels: The reduction in search time expressed as per cent gain varied between 30% and 70%. A modest computation of quantitative advantage derived through the usage of PFZ advisories along the Kerala coast for different types of vessels considering an average of 50% reduction in search time indicates that annual savings on account of diesel consumption for mechanised vessels is about Rs.6.0 lakhs, large motorised boats is Rs.1.95 lakhs and small motorised boats is Rs. 1.65 lakhs. Considering that 27 % of the boat owners are using PFZ advisories this amounts to a saving of Rs.163 crores for Kerala coast. If 100 % of the mechanized and motorized boats operating in Kerala use PFZ advisories, this will account for annual savings of about Rs.600 Crores just on account of diesel savings in addition to the valuable human effort. Extrapolation of the above results to the national scenario indicates a savings of Rs. 1351 crores for 27 % usage and Rs. 5000 crores for 100% usage. Putting it in a different perspective, the national consumption of diesel on this count would reduce by 166 Cr litres per annum. This will go a long way in even reducing environmental pollution.

CPUE and Cost Benefit analysis for Ring Seining: From the controlled experiments conducted to validate 18 PFZ advisories, generated by INCOIS, results clearly indicate the effectiveness and usefulness of PFZ advisories to enhance the economics of fishing operations. All 18 experiments resulted in better catch and income for vessels fishing in notified PFZ locations as compared to those fishing in non-notified locations. The increase in CPUE for vessels using PFZ advisories is about 4.3 times more than those vessels fishing in non-notified locations. The corresponding increase in income is about 4.8 times for vessels fishing in notified fishing locations as compared to those fishing in non-notified location. 100% success rate in PFZ locations could be attributed to the fact that all the 18 experiments targeted pelagic resources using ring-seiners re-emphasizing the fact that PFZ advisories are very effective for pelagic fishing resources.

Percentage Success of Fishing Operations for different types of fishing operations: Percentage success of fishing operations in the notified PFZ locations was 92% for ring seining, 95% for gill netting and 85% for bottom trawling operations as compared to a meager 29%, 12% and 0% respectively in the non-notified locations. These experiments conclusively prove the effectiveness of using Potential Fishing Zone advisories. To sum up, where fishing is targeted at a pelagic resource which is migratory in habit, PFZ will definitely provide an advantage in terms of reduction in searching time, in turn reduction in the consumption of fuel oil and human effort which ultimately will bring down the total cost of fishing operations. It is conclusively proven that there is a certain improvement in CPUE and Net Profit due to the use of PFZ advisories. The percentage success rate of PFZ advisories is about 95% for pelagic operations such as ring seining and gill netting and 85% for bottom trawling operations.

While considering the above results, it is pertinent to note that the study was taken up only for the Kerala coast and an attempt has been made to extrapolate to obtain the national scenario. The national estimates are just indicative and similar detailed investigations need to be taken up along other parts of the coastline to get the best estimates. The percentage of success in the notified locations can be improved by addition of other important parameters such as wind, mixed layer depth, etc in addition to seas surface temperature and chlorophyll. Increasing awareness among the fishing community is very important for wide use of this technology. INCOIS has already initiated efforts in this direction.

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